SCORE: Synthesising Complete, Owned, Responsive Environments for Simulation

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Abstract

Games, film, robotics and embodied AI need 3D environments in which every object is a physical body of its own, that stay editable, and that belong to whoever made them. World models produce video that no simulator can open; one-shot 3D generators produce a single fused scene. SCORE is an open framework that synthesises such environments the way studios build them in OpenUSD, scene within scene and layer by layer, every output passing measured tests before the next is made. Every person in an environment also receives a musculoskeletal body that follows its motion in simulation. This opens a route to training data that recorded measurement cannot supply at scale: movement in complete scenes, with physics-derived ground truth down to the forces of individual muscles. Code: https://github.com/Babon-Innovations-b-v/score (MIT).

Gap: Figure 1, one environment, every layer. Environment 1 from the top down: its physics (the bodies it stands on and their gravity), a top scene on its site, a parent scene with its child scenes, and one scene opened into its layers side by side (base, settle, physics, light, motion, effects, sound, people). It waits on environment 1's batch.

1 Introduction

Simulated 3D environments serve games, film, robot learning and embodied AI [1]. Across these uses they share a common structure of separate objects with collision geometry, mass and joints, a ground surface with its gravity, light and sound emitted by identifiable sources, and scenes that can be reopened, modified and simulated again. Authoring such an environment by hand is expensive, and a generative system that takes over this work must leave the result in the hands of its author, since in the Game Developers Conference's 2026 survey 52 percent of game workers considered generative AI to be harming the industry [2].

World models generate an environment as a sequence of video frames [3], [4] that no physics simulator can step forward in time. 3D environment generators [5], [6] produce geometry, mostly as a single fused asset from which separate objects have to be recovered afterwards [7], [8]. Both operate in a single pass, so a fault in the output cannot be traced to the step that caused it. Procedural content generation systems [9], [10] and coding agents, language models that write scenes as executable programs [11], [12], [13], [14], [15], [16], keep the environment explicit and inspectable, but they either reconstruct a scene that already exists or assemble it from a fixed asset library. The models they produce often float above their supports or come apart [17], such agents assess their own geometry unreliably [18], and on scenes reconstructed from video the strongest model scores 0.91 on static scenes but only 0.67 on dynamic ones [19]. Open vision foundation models [20], large models pretrained on broad image data and reused across tasks, now recover 3D shape [21], segmentation

masks [22] and depth [23] from images, and a coding agent can call them as tools, yet none of them gives a consistent visual style across objects that different models produced [24], [25], and none gives physical properties.

The components needed for the physics already exist. OpenUSD [26], the scene description that Pixar developed for its film production, stores each department's work in its own layer file and composes scenes from smaller ones by reference, and its physics schema describes rigid bodies, colliders, mass and joints within the same stage, the composed scene. MuJoCo [27] is a widely used physics engine for articulated bodies in contact, and Newton [28], built on the Warp GPU programming framework [29], runs MuJoCo's solver on the GPU [30] and loads OpenUSD scenes directly. What is missing is a route from what a creator intends to a complete environment in this form.

SCORE is that framework. Its environments are complete, since every object, surface, light and sound a scene shows exists wherever the viewer looks. They are owned, because every model in the framework permits commercial use of its output, so the creator holds the full rights. They are responsive, because every object carries its collider, mass and material and every moving part its joints, so a simulator steps the environment as written. They are editable, since each scene is a stack of layers produced by steps that are each identified by a key derived from their inputs, and a change reruns only the steps it affects. An environment can range from a house on a plot of ground to a system of worlds that move one another by gravitation.

We contribute the SCORE framework, which turns a creator's locked design into a single OpenUSD environment that is physically simulable and opens unchanged in a game engine, a robot simulator or a film renderer. Its method (Section 2) builds a scene from the top down and layer by layer, with each stage passing measured acceptance tests before the next begins. The evidence plan (Section 3) comprises four demonstrations on a first environment, left as gaps until its batch has run. Section 4 sets out where the framework goes next, towards simulated people whose bodies are true to the individual person.

2 Method: building an environment

2.1 From concept to plan

The environment concept. The creator first locks the EC, which fixes the environment's visual style and physical setting together with his reference images and the scenes it contains, in a short structured interview conducted with the vision-language model Gemini 3.1 Pro [31]. An environment plan then places the site of each top scene on that ground and computes the sky visible from it.

A concept for each scene, top down. A concept image is then locked for each scene, parents before their child scenes. Each candidate image is one call to the image generation model Nano Banana Pro [32], made on a concept review page where the creator marks regions and requests a new image. A child scene is drawn only once its parent has been locked and planned, from a crop of the parent's concept, so that it fits the parent it stands in.

Inventory and plan. Gemini 3.1 Pro lists each object of the locked concept as a row of the scene inventory, giving its kind, count, anchor and size in metres, and the segmentation model SAM 3 [22] localises each row with a bounding box. Nothing outside the inventory is generated or placed. Nano Banana Pro then draws the scene from directly above, and Gemini 3.1 Pro writes the plan in metres from that drawing, the concept and the inventory. The plan gives the outline, walls,

openings, levels, stairs, roofs and terrain, each item with its orientation, people as the spots on which they stand, and the rooms of the child scenes. Free checks, which make no paid model call, test that every row is placed as many times as its count, inside the outline and clear of other floor items, that every door and standing spot can be reached on foot, and that the child scenes tile their parent. Any remaining violations are returned to the model as notes at most twice, and only what still fails is recorded as a gap. The creator's edits on the plan review page always take precedence and cause everything that follows to be recomputed.

Shapes first. The route follows one coarse-to-fine loop at every scale. A parent scene receives its concept, plan and a rough block model, and its plan is divided into the shapes of its child scenes. Once those shapes are fixed, all child scenes are produced in parallel, each repeating the loop. The stages `rough-model` and `shell-split` are implemented but have not yet run.

2.2 Layers from row kinds

A scene's layers are derived from its design alone. Each inventory row's kind, determined from the noun in the inventory reading by a single table of rules, states what its objects do, whether they emit, move, lie loose, glow, grow, act or sound. A scene receives an effects, motion, physics, crops or people layer only for rows that do so, and always a light and a sound layer. Each layer is generated from its rows by the rule of their kind, so a new environment needs no hand-written list of layers. The scene record holds what no generated object covers (building shells, stairs, ground and water, the distant backdrop, lamps and the neighbouring scenes its doorways lead to), and its gate requires the shell to lie within 5 cm of the plan and every doorway to open onto a floor.

Models. Objects follow the prop route first, which turns a single image of an object into a 3D model. Nano Banana Pro draws a close-up of the object from its crop of the concept and redraws it at most three times when the shape check rejects it. The image-to-3D model Pixal3D [21] then reconstructs the mesh, and a likeness gate compares its silhouettes from eight viewpoints with the close-up. Only an object whose checks fail is built instead by a procedural builder, a program that a coding agent writes, and a thin straight object (a pipe, post or beam) whose every close-up failed is generated in code by sweeping its cross-section along its length.

Surfaces. Everything a scene shows comes from the creator's asset library, score_lib, or is recorded as a written exception. A generated model contributes geometry only, and each of its parts takes one material from a library of procedural materials, written as ProcFunc functions [33] on Infinigen's shaders [9] and coloured from the scene's palette. The finish route labels each face of the model from free renders, matches each finish to a library material, bakes it into textures and checks the bake against the close-up, requiring each finish to lie within a colour difference of ΔE 32 of its colour there. A free library check traces every material, model, light, sound and character of an accepted assembly back to the library or to an exception.

Settle and physics. Loose objects are dropped onto the generated ground in a rigid-body simulation, the settle, and an object that would rotate by more than 15 degrees or drift by more than 30 cm is recorded as a resting fault and keeps its laid-out pose. The settle writes the pose in which each object comes to rest as a layer. The physics trial then steps the composed scene in Newton, using MuJoCo's solver with Newton's contact handling under the gravity of the body the scene stands on, and evaluates it with the gates `reacted`, `at rest`, `energy` (no increase in the energy of undriven bodies) and `hinge` (a hinged door leaf swings to its joint limit). The trial moved from Blender's Bullet engine [34] to Newton on 27 October 2026. On a test scene at lunar gravity (two dropped crates, a driven door with a wheel fixed to it and a pushed stool), both engines gave the

same result on every gate, and Newton brought the crates to rest on their targets where Bullet left them standing off by its contact margin of 8 mm. Newton's deterministic mode, which reproduced bitwise-identical results over six runs of that test scene, cannot handle scenes with many loose bodies, because its buffers exceed the 32-bit limit, so the trial and the settle run without it. Two runs of a 25-body test scene then agree within 5e-7 m for every body except one that falls off the edge of the floor. The settle has run in Newton in the same way since 27 October, so one engine steps both what comes to rest and what is tested; one effect is to be simulated in Newton as a first case for the effects layer.

Light, motion, effects and sound. The sky is computed from the physical setting in the EC, never drawn, with each celestial body at its true direction, apparent size and phase as seen from the scene's site. The lamps are the inventory's glowing rows. One render per scene places each lamp in its own light group, and a free sweep mixes their intensities to bring the scene's brightness within a factor of 1.5 of its concept's. Every object of a moving row moves by the rule of its kind at a speed taken from cited real machines, and is gated on nothing coming within 3 cm of anything else. Emitters are measured against real footage. Each sound source receives a library file or one take generated by the text-to-audio model MOSS-SoundEffect [35], gated on an audio-text similarity score from CLAP [36] of at least 0.30.

People. People are models like any other, each made from a close-up per person row, reconstructed by Pixal3D and given its materials by the finish route. The SOMA parametric body model [37], estimated from the close-up by SAM 3D Body [38], is fitted beneath the model as its skeletal rig and animated with motion clips from Kimodo [39]. A shell gate accepts a clothed appearance when 99 percent of the points of its non-rigid parts lie within 5 cm of the body. Each person stands on a person spot of its scene's plan.

Sim bodies. Each character also receives a sim body, a musculoskeletal twin that follows its movement exactly while Newton and MuJoCo supply the physics. The body is MyoFullBody [40], whose arm and hand derive from the MoBL-ARMS model [41], [42], with 416 muscles, 86 bones and 95 independent joints, 36 more joints being coupled to others. Its bone groups are sized to the person's SOMA body, and its masses are spread by SOMA's segment volumes. One solver then fits it to the clip: the person's constants once, and every frame's joint angles against each SOMA segment's rotation, 84 of SCORE's 102 anatomical landmarks on the SOMA skin and SOMA's joints at the ends of each bone group, with MyoFullBody's joint couplings held exactly and a cost that keeps the bones inside the skin. SOMA is never moved or reshaped, except that each clip's standing feet are put on the floor by raising or lowering the whole clip. For every frame, inverse dynamics on the scene's own ground then gives the forces the movement needs, split in the two stages of OpenSim's static optimisation [43]: the ground reaction force first, from the pelvis' balance alone and shared over the points of skin touching the ground, then the muscle activations with that force fixed. A residual reduction in the manner of OpenSim's RRA adjusts the pelvis, trunk and masses within a per-frame cap, and whatever force remains at the pelvis is recorded as the residual and never applied. Solving contact and muscles together was ill-posed, with a nudge of 1e-6 to the contact columns moving an activation by up to 1.0; in two stages the same nudge moves one by at most 5e-3.

On main, a sim body follows SOMA closely: on a 6 s jump 1.37 percent of the bone lies outside the skin, the median joint-centre error is 7.0 mm and the median segment rotation error 4.4 degrees, and on a standing clip 1.14 percent and 7.4 mm. All of it runs on one GPU. Physics, the residual reduction and its foot holds run every frame at once on the card with MuJoCo Warp, as do the per-frame pose fit and the posing of the SOMA skin, and the card's pose fit equals the CPU reference

(mink) within 0.01 degrees and 0.01 mm on every frame. A 6 s jump (180 frames at 30 fps) now takes 120 s on one NVIDIA L4 and a 5 s standing clip 89 s, against 830 s and 412 s a day earlier, plus about 3 min for a fresh card's first kernel compile. The card's stages run in Warp's deterministic mode, so two runs give the same result bit for bit.

A walk's or a lunge's two standing feet stood at different heights, because a person's legs differ from those of the skeleton the motion generator moved. Kimodo's own foot-contact correction therefore runs after this on the person's own legs, with the generator's own contact labels, and bends a toe back toward flat where its pin left the foot under the floor. On the contact frames of a walk and a lunge the median sole then stands within 0.2 mm of the floor, the frames on which the ground carries nothing go from 5 to 0, and the largest residual force falls from 782 N to 57 N on the walk and from 531 N to 227 N on the lunge.

Gap: Newton for the scenes, one effect and the sim body validated. The resting and physics gates per scene of environment 1 with the settle and the trial on Newton; the one effect stepped on Newton; and a sim body's joint angles, ground reaction forces and joint moments against OpenCap's LabValidation motion capture and force plates [44]. It waits on environment 1's batch, the effect's stage and a validation run.

Completion. A scene is complete when its completion gate passes on its final assembly, with every inventory row made or dropped with a reason, nothing left unmade, no resting fault, and every layer and every visible scene present. A fault the creator chooses to ship with is an accepted gap, recorded with who decided, when and why. The gate lists every gap by name and is never relaxed. A hook prevents a coding agent from claiming that a scene is done while its gate fails, because agents judging their own scenes agree with a measured geometry score only 45.8 percent of the time [18]. The films of a complete scene are rendered from scripts the creator has locked.

Buying in phases. A paid run, a Buy, runs as phases (models, surfaces, people, light and sound, assembly, film) rather than in one pass. Before a phase is queued, a free preflight plans it, imports its code in a fresh process and runs one node of each paid stage up to the point where it would rent a machine; once queued, its code and inputs are fixed. Each output that passes its checks is locked, so a later code change never makes it again, and a failing one gets one retry before it becomes a gap. Paid jobs rent cloud machines or run on a shared Kubernetes cluster, chosen per kind of job; the image-to-3D, Blender, Newton settle and motion-clip jobs have run there.

3 Evidence: environment 1

Environment 1 serves as the evidence. The creator did not accept its first run, and its store of outputs was lost on 19 October 2026, so this draft shows none of its outputs, and each demonstration below will be produced by its new batch. Figure 1 is the first.

Gap: Demonstration 2, physics that holds. Environment 1's scenes stepped on Newton under their own body's gravity: loose objects come to rest, doors swing on their hinges, with the physics trial's gates per scene and Bullet run on the same scenes as the baseline. It waits on environment 1's batch and on Newton taking over the settle.

Gap: Demonstration 3, one environment, many uses. The same OpenUSD environment opened unchanged in a game engine, in a robot simulator and in a film render, with what each reads of its layers and what it drops. It waits on environment 1's batch and on the engine adapter.

Gap: Demonstration 4, an edit that stays cheap. The creator changes one thing in a locked scene, and only the stages that the change affects run again; the stages rerun, time and euros against a full rebuild. It waits on environment 1's batch and on the edit layer being carried from one assembly to the next.

Gap: Table 1. Per scene of environment 1: inventory rows, models generated and code-built, layers, gaps before and after the batch, time, euros and the creator's verdict. It waits on environment 1's batch and the creator's review.

Disclosure. The creator's locks, picks, accepts and waivers are his own actions in the cockpit, and everything between them is automatic. Coding agents wrote most of the framework's code under the author's direction, and 3,281 of its 3,306 commits list a coding agent as co-author. Manual fixes to a scene are recorded as the creator's edits.

4 Future work: people true to the person

This section describes goals, not results.

Bodies true to the person. The sim body is sized to the person but still built on one generic model, and it is to become a musculoskeletal model of the individual person, including older people and people with a curved spine. Its joint ranges and couplings are to become the person's own, read from that person's movement and starting from the published healthy ranges, with any widening flagged for review. Every input that SOMA does not provide, such as muscle strength and bone shape, is marked as assumed and given a plausible range. The joint angles, and then the net joint moments and ground reaction forces, are to be validated against OpenCap's LabValidation data.

Scenes from motion. Motion comes from the SOMA ecosystem: estimated from video with GEM-X [45], or generated from text and constraints with Kimodo [39], in practically unlimited quantity. The terrain a movement implies becomes a SCORE scene, so the movement keeps its setting without the source video or anything that identifies a person. The sim body follows each movement on the scene's own ground. For every frame, the simulation solves for the ground reaction forces from contact, the net joint moments, and the activations and forces of the muscles that reproduce the movement. Whatever force and moment neither the ground nor the muscles can supply remains at the pelvis as the residual. The residual is recorded and never applied to the body, so it measures how dynamically consistent the movement is, and only movements with a small residual are kept. Biomechanics uses the same residual to assess recorded motion [43], [46], and PhysibleMDM [47] minimises an inverse-dynamics residual to make generated motion feasible for a torque-driven robot. We apply the criterion to generated motion in a muscle-driven body.

Layered training sets. The same simulations yield training sets in layers. Each sample pairs a rendered image of a clothed person in a complete scene with ground truth for every layer beneath it: the SOMA body and its joint angles, the ground reaction force under each foot, the muscle activations and forces, the residual, and the depth, segmentation and layout of the scene. Their main use is to train estimators that infer muscle activations, joint moments and ground reaction forces directly from ordinary video, fast enough for routine clinical use. Recorded data cannot supply this at scale: muscle forces cannot be measured without invasive methods, and force plates and electromyography confine measurement to a laboratory. Simulation can generate the bodies, movements and settings an estimator must cover, including those of a specific patient group. BioHuman10M [48] already pairs video, part of it rendered, with the activations of a 112-muscle model and with ground reaction forces predicted by a learned model. Our sets would add scenes generated for the purpose under varied lighting, terrain and clothing, ground reaction forces from contact

with each scene's own ground, a 416-muscle model, and a residual for every sample. Their value depends on whether estimators trained on them transfer to real force-plate and electromyography recordings.

Brains. A brain is a trained controller that drives a sim body's muscles. Trained on the retained movements and the muscle forces that produce them, it is placed in SCORE's synthesised worlds under their own gravity and terrain, where it must adapt to situations no recording contains. Every run it drives adds movement to the training sets, so it extends the data beyond what a motion generator produces. Muscle-driven controllers such as KINESIS [49] and MuscleMimic [50] are trained on motion capture, as are torque-driven ones such as PHC [51]. To our knowledge, no muscle-driven controller has yet been trained on generated motion selected by its residual. That claim holds only once such a controller is compared with one trained on motion capture and with one trained on the same generated motion unfiltered.

The data loop. The amount of data scales with the compute put behind it, since every scene, movement and run adds samples. This is why SCORE is open: others can apply their own compute to the same framework and share the resulting datasets.

5 Limitations and conclusion

A single creator judges the results, and he has not yet accepted a run. Closed models (Nano Banana Pro, Gemini 3.1 Pro) draw the concepts and close-ups and read and plan every scene, and may change or be withdrawn. The design step, the layers derived from row kinds, the finish route, the shapes-first loop and the scene tree have not yet carried a scene through a batch. The edit layer is so far only a reserved slot. The effects do not yet run in Newton, the sim body has not been validated against measured forces, and no game engine or robot simulator has yet opened the result. Next come environment 1's batch with its four demonstrations, an engine adapter, the edit layer carried across assemblies, further environments, open models replacing the closed ones, and the data loop, in which residual-filtered generated movement trains muscle-driven controllers whose value a comparison with motion capture has yet to show. SCORE synthesises an environment as studios build one, scene by scene and layer by layer, in a form a simulator steps as written, and an accepted environment will show whether the approach holds.

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